

Financial Econometrics I

High-frequency financial models - HAR, Realized GARCH (Lecture 9)

Winter Semester 2022/2023 by Jozef Baruník

* If viewed in .pdf format - for full functionality use *Mathematica 7* notebook (.nb) version of this .pdf

Outline

We already know how to utilize high-frequency data to compute Realized Volatility and Jumps.

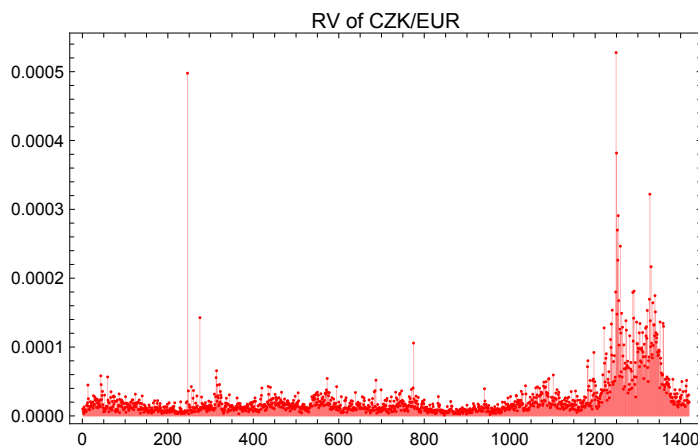
Today, we will learn how to make use of the realized measures.

- build a forecasting model with realized measures - ARFIMA, HAR, Realized GARCH
- Value-at-Risk with HF measures

How to build a model on Realized Volatility?

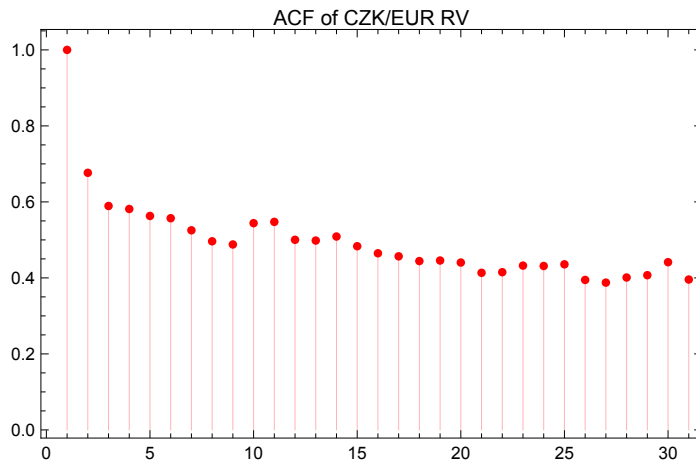
Code

Example of CZK/EUR volatility



Example of CZK/EUR volatility structures

Let's compute ACF of CZK/EUR 5-minute Realized Variance for the period of 2004-6/2009.



How would you model series with such structure?

Heterogeneous Autoregressive (HAR) model

A good idea is to try AR structures although we may suspect long memory

Heterogeneous Autoregression (HAR) of F.Corsi (2009):

$$RV_t = \alpha_0 + \beta_1 RV_{t-1} + \beta_2 RV_{t-1}^{(5)} + \beta_3 RV_{t-1}^{(22)} + u_t,$$

where $RV_{t-1}^{(h)}$ is h -period realized variance, i.e. $RV_{t-1}^{(5)}$ corresponding to 1 week and $RV_{t-1}^{(22)}$ corresponding to 1 month, and u_t normally distributed error term.

The model is based on the idea of heterogeneous investment horizons of traders, i.e. 1-day, 5-days (week) and 22-days (1 month)

We can also consider volatility and logarithmic version of HAR:

$$\sqrt{RV_t} = \alpha_0 + \beta_1 \sqrt{RV_{t-1}} + \beta_2 \sqrt{RV_{t-1}^{(5)}} + \beta_3 \sqrt{RV_{t-1}^{(22)}} + u_t,$$

$$\ln(\sqrt{RV_t}) = \alpha_0 + \beta_1 \ln(\sqrt{RV_{t-1}}) + \beta_2 \ln(\sqrt{RV_{t-1}^{(5)}}) + \beta_3 \ln(\sqrt{RV_{t-1}^{(22)}}) + u_t,$$

An alternative is to use ARFIMA for RV due to its long memory features. HAR is only approximate long memory model

Realized GARCH

A good idea is also to use power of GARCH and explore HF data in one framework.

Recall GARCH has three components

$$\mu_t = E(r_t | \mathcal{F}_{t-1})$$

$$h_t = \text{var}(r_t | \mathcal{F}_{t-1})$$

$$z_t = \frac{r_t - \mu_t}{\sqrt{h_t}} \sim F$$

where key is to model $h_t = g(\mathcal{F}_{t-1})$, usually $\mu_t = \mu$ is constant and $F \sim \text{iid } N(0,1)$

GARCH

Standard GARCH(1,1) of Bollerslev (1986)

$$h_t = \omega + \alpha r_{t-1}^2 + \beta h_{t-1}$$

Estimation

Use likelihood for time series models, with key objective log-density for observed data $f_\theta(r_t | \mathcal{F}_{t-1})$ where θ is full set of parameters.

In case of conditional return distribution $r_t | \mathcal{F}_{t-1} \sim N(\mu, h_t)$, likelihood function is

$$L(\mu, \omega, \alpha, \beta) = \prod_{t=1}^T \frac{1}{\sqrt{2\pi h_t}} \exp \left\{ -\frac{1}{2} \frac{(r_t - \mu)^2}{h_t} \right\}$$

and log-likelihood

$$\text{LogL}(\mu, \omega, \alpha, \beta) \propto -\frac{1}{2} \sum_{t=1}^T \left(\log(2\pi) + \log h_t + \frac{(r_t - \mu)^2}{h_t} \right)$$

that we maximize, inference done by Score and Hessian

GARCH Zoo

ARCH:	$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i u_{t-i}^2$
GARCH:	$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i u_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$
IGARCH	$\sigma_t^2 = \omega + u_{t-1}^2 + \sum_{i=2}^p \alpha_i (u_{t-i}^2 - u_{t-1}^2) + \sum_{j=1}^q \beta_j (\sigma_{t-j}^2 - u_{t-1}^2)$
Taylor/Schwert:	$\sigma_t = \omega + \sum_{i=1}^p \alpha_i u_{t-i} + \sum_{j=1}^q \beta_j \sigma_{t-j}$
A-GARCH:	$\sigma_t^2 = \omega + \sum_{i=1}^p [\alpha_i u_{t-i}^2 + \gamma_i u_{t-i}] + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$
NA-GARCH:	$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i (u_{t-i} + \gamma_i \sigma_{t-i})^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$
V-GARCH:	$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i (e_{t-i} + \gamma_i)^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$
Thr.-GARCH:	$\sigma_t = \omega + \sum_{i=1}^p \alpha_i [(1 - \gamma_i) u_{t-i}^+ - (1 + \gamma_i) u_{t-i}^-] + \sum_{j=1}^q \beta_j \sigma_{t-j}$
GJR model:	$\sigma_t^2 = \omega + \sum_{i=1}^p \left[\alpha_i + \gamma_i I_{\{u_{t-i}^2 > 0\}} \right] u_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$
NGARCH:	$\sigma_t^\delta = \omega + \sum_{i=1}^p \alpha_i u_{t-i} ^\delta + \sum_{j=1}^q \beta_j \sigma_{t-j}^\delta$
log-GARCH:	$\log(\sigma_t) = \omega + \sum_{i=1}^p \alpha_i e_{t-i} + \sum_{j=1}^q \beta_j \log(\sigma_{t-j})$

$$\text{EGARCH: } \log(\sigma_t^2) = \omega + \sum_{i=1}^p [\alpha_i e_{t-i} + \gamma_i (|e_{t-i}| - E|e_{t-i}|)] + \sum_{j=1}^q \beta_j \log(\sigma_{t-j}^2)$$

$$\text{AP-GARCH: } \sigma_t^\delta = \omega + \sum_{i=1}^p \alpha_i [|u_{t-i}| - \gamma_i u_{t-i}]^\delta + \sum_{j=1}^q \beta_j \sigma_{t-j}^\delta$$

$$\text{H-GARCH: } \sigma_t^\delta = \omega + \sum_{i=1}^p \alpha_i \delta \sigma_{t-i}^\delta [|e_t - \kappa| - \tau(e_t - \kappa)]^\nu + \sum_{j=1}^q \beta_j \sigma_{t-j}^\delta$$

$$\text{GQ-ARCH: } \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i u_{t-i} + \sum_{i=1}^p \alpha_{ii} u_{t-i}^2 + \sum_{i < j}^p \alpha_{ij} u_{t-i} u_{t-j} + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$$

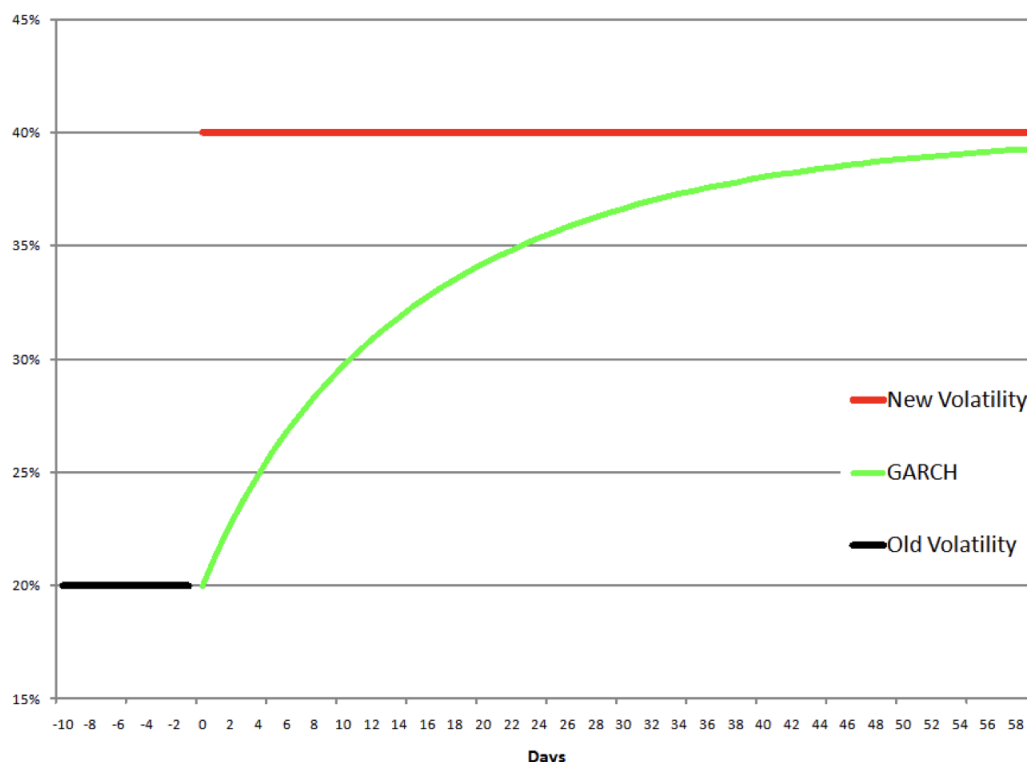
$$\text{Aug } \sigma_t^2 = \begin{cases} |\delta \phi_t - \delta + 1|^{1/\delta} & \text{if } \delta \neq 0 \\ \exp(\phi_t - 1) & \text{if } \delta = 0 \end{cases}$$

$$\begin{aligned} \text{-GARCH: } \phi_t = & \omega + \sum_{i=1}^p [\alpha_{1i} |u_{t-i} - \kappa|^\nu + \alpha_{2i} \max(0, \kappa - u_{t-i})^\nu] \phi_{t-j} \\ & + \sum_{i=1}^p \left[\alpha_{3i} f(|u_{t-i} - \kappa|, \nu) + \alpha_{4i} \frac{\max(0, \kappa - u_{t-i})^\nu - 1}{\nu} \right] \phi_{t-j} \\ & + \sum_{j=1}^q \beta_j \phi_{t-j}^2 \end{aligned}$$

GARCH is updating slowly

conventional GARCH take information from returns to adjust value of conditional variance

Can high frequency data help?

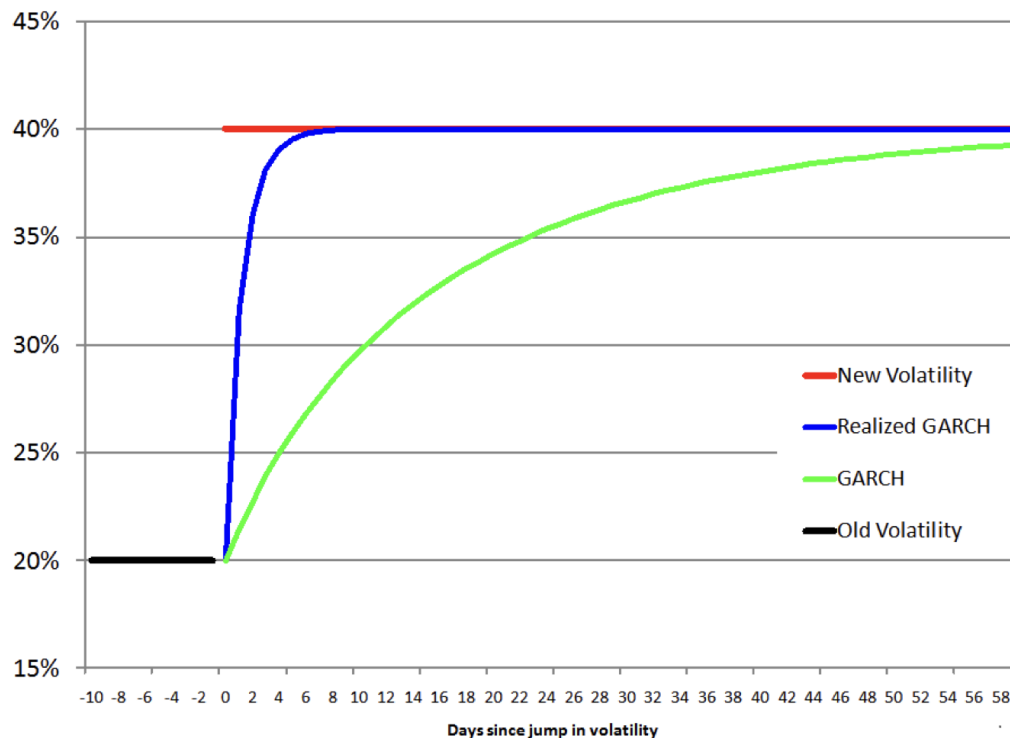


Realized GARCH

While volatility is latent in GARCH model, we can think about “helping” GARCH by including realized volatility into the volatility equation. For this, we need so-called measurement equation, as realized volatility is measured with some error. General idea is:

$$h_t = \omega + \alpha r_{t-1}^2 + \beta h_{t-1} + \gamma x_{t-1}$$

where x_t is realized measure of volatility and hugely improves fit



But we need measurement equation for RV since this is measured from data with noisy non-parametric estimation.

Since the data contain jumps, logarithmic specification is also preferred

Complete Realized GARCH then takes form

$$\begin{aligned} r_t &= \sqrt{h_t} z_t \\ \ln(h_t) &= \omega + \beta \ln(h_{t-1}) + \gamma \ln(x_{t-1}) \\ \ln(x_t) &= \xi + \psi \ln(h_t) + u_t \end{aligned}$$

where the first equation models returns as normally distributed z_t term with volatility $\sqrt{h_t}$, which is modelled by second equation. Realized volatility helps second equation, but is also measured with some noise u_t , so we need to add third equation to connect latent variance h_t to RV_t .

Estimation of parameters is done via QMLE. But in comparison to GARCH, we now have two random variables in the model, z_t and u_t , so the likelihood function will be product of the pdfs of these two variables.

Estimation - QMLE

Gaussian errors $z_t \sim \text{iid } N(0, 1)$, and $u_t \sim \text{iid } N(0, 1)$

a quasi log-likelihood to be maximized is

$$\text{LogL}(r, x; \theta) = -\frac{1}{2} \sum_{t=1}^T \left(\log h_t + \frac{r_t^2}{h_t} + \log \sigma_u^2 + \frac{u_t^2}{\sigma_u^2} \right)$$

that we maximize, inference done by Score and Hessian

Value-at-Risk (VaR) using HF data

Prominent problems in finance calling for understanding entire distribution of returns are

- risk management and measurement (VaR)
- portfolio allocation when returns are non-Gaussian
- hedging and market timing strategies
- derivative pricing

These all require full characterisation of the multi-period forecasts of return distribution.

Until now, we have only learned how to forecast variance of returns (second moment). The question is, can we use realized measures to model distribution of returns?

Value-at-Risk example (VaR)

VaR is defined as the maximum expected loss which may be incurred by a portfolio over some horizon with given probability. Let $q_\alpha(r_{t+1})$ denote the α -quantile of the return distribution. VaR_α of a given portfolio is then simply

$$q_\alpha(r_{t+1}) = \sigma_t \phi(\alpha)$$

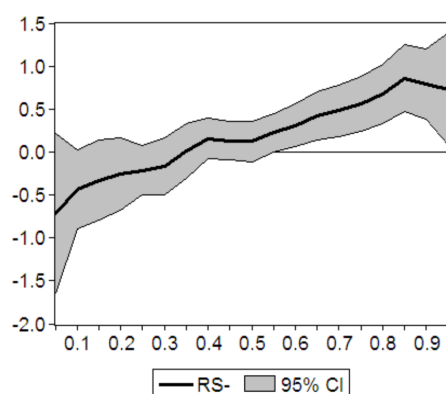
where $\phi(\alpha)$ is α -quantile of a standard normal cumulative distribution, and σ_t estimated volatility. We can use Realized volatility here. For example using HAR, we can obtain future forecasts of VaR.

Still, this approach is restrictive, as it requires assumption on distribution. One can also think about quantile regressing future returns on realized volatility as:

$$q_\alpha(r_{t+1}) = \beta_0(\alpha) + \beta_{RV}(\alpha)' RV_t$$

where we can use various realized measures as volatility, semivariance, jumps to forecast quantiles of returns.

The quantile regression turns to be a simple idea, which does not require any assumptions, and provide by definition conditional forecast of quantiles. One example, how negative volatility drives different quantiles of returns:



(Optional) Reading:

Zikes, F. and Barunik, J. (2016): Semiparametric Conditional Quantile Models for Financial Returns and Realized Volatility, Journal of Financial Econometrics, forthcoming

<http://jfec.oxfordjournals.org/content/early/2014/10/21/jjfinec.nbu029.short?rss=1>

or

<https://ideas.repec.org/p/arx/papers/1308.4276.html>

Next Week

Distributional departures and asymmetries

- jumps in forecasting
- higher order moments
- asymmetries